University of Sri Jayewardenepura, Sri Lanka

Bachelor of Information and Communication Technology Semester 5

ITM3232 Image Processing

Laboratory Exercise 2

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Image resizing is an important operation in digital image processing. It is used to increase or decrease the dimensions of an image while maintaining image quality as much as possible. Different interpolation methods produce different image qualities during resizing.

following Python libraries were used in this laboratory exercise:

- OpenCV (cv2)
- Matplotlib (matplotlib.pyplot)

Displaying the Original Image

1. INTER_NEAREST

INTER_NEAREST is the simplest and fastest interpolation method. It selects the value of the nearest neighboring pixel without considering surrounding pixels.

- When enlarging the image appeared blocky and pixelated.

- Fine details were not preserved properly.
- During downscaling the image quality decreased.

This method is computationally fast but produces lower quality images during enlargement.

2. INTER_LINEAR

INTER_LINEAR uses the weighted average of neighboring pixels to calculate new pixel values.

- Produced smoother images compared to INTER_NEAREST.
- Reduced pixelation during enlargement.
- Maintain moderate image quality for both enlargement and shrinking.

This method provides balanced image quality and processing speed.

3. INTER_CUBIC

INTER_CUBIC uses cubic interpolation based on nearby pixels and provides smoother transitions between pixels.

- Generate sharper and smoother enlarged images.
- Preserved image details better than INTER_LINEAR.
- Produced high quality results at scale enlargement.

This method improved image quality significantly but required more processing time.

4. INTER_LANCZOS4

INTER_LANCZOS4 uses a high quality Lanczos algorithm with a larger pixel neighborhood.

- Produced the highest quality enlarged images among all methods.
- Preserved fine image details and edges effectively.
- Images appears smoother and clearer compared to other methods.

This method generated excellent image quality. But slower than other interpolation methods.

5. INTER_AREA

INTER_AREA is mainly used for image shrinking and works by resampling pixel areas.

- Produce the best results during image reduction.
- Reduced aliasing and preserved image quality while shrinking.
- Enlargement quality was not as good as other methods.

This method is most suitable for downscaling images.